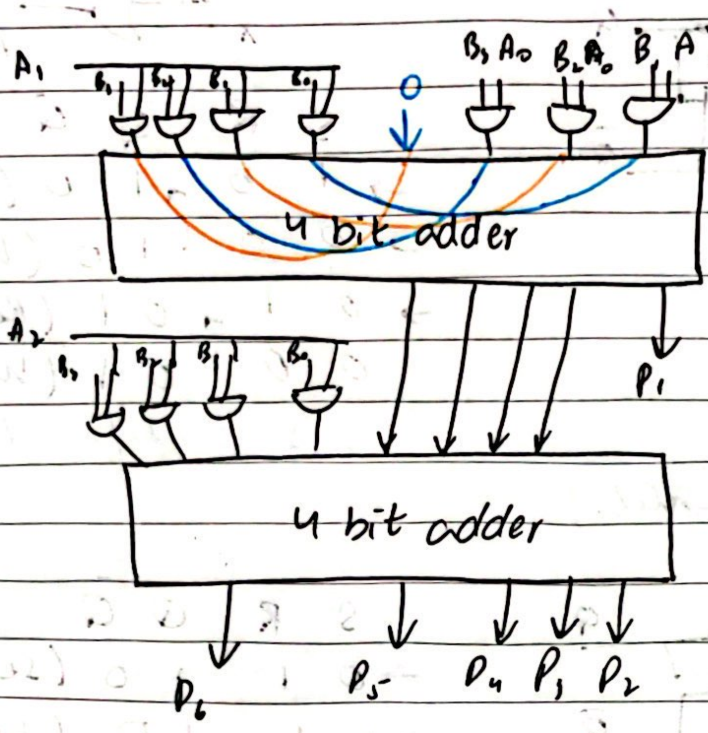
Date

A_2B_3 A_2B_2 A_2B_1 A_2B_0 X X

P_6 P_5 P_4 P_3 P_2 P_1 0

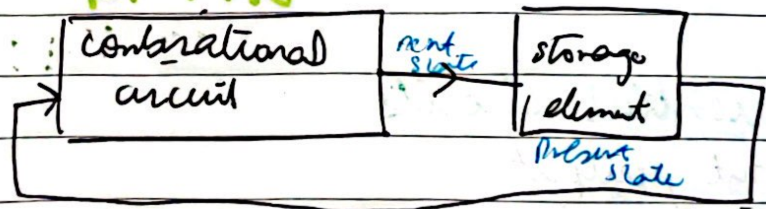


DAY-18 (After mid-2)

Sequential Circuits :-

Automatic traffic system

PI-VAD



→ state of the system present state, Next state

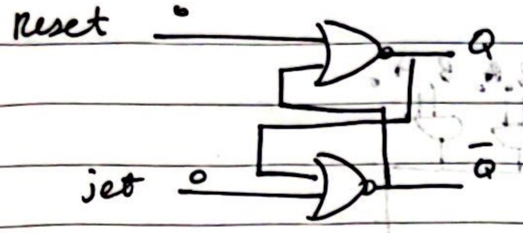
Basic Building block:.

latch:.

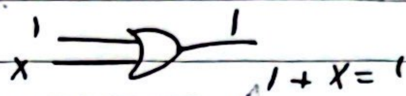
stage element that can maintain binary state infinitely as long as power is on,

Date

SR latch (NOR Latch)



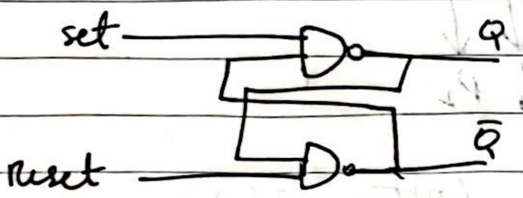
→ 1 - dominance



set
Reset
hold.

S	R	Q	$\bar{Q}$	
1	0	1	0	(set)
0	0	1	0	(hold)
0	1	0	1	(Reset)
0	0	0	1	(hold)
1	1			undefined

$\bar{S}\bar{R}$ latch (NAND latch)



S	R	Q	$\bar{Q}$	
0	1	1	0	(set)
1	1	1	0	(hold)
1	0	0	1	(Reset)
1	1	0	1	(hold)
0	0			undefined

→ low enserted

DAY-19

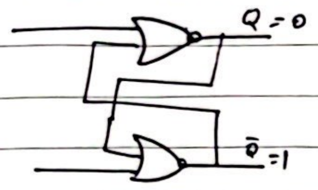
Sequential circuits

→ Input, output
→ state variables
→ present state, next state

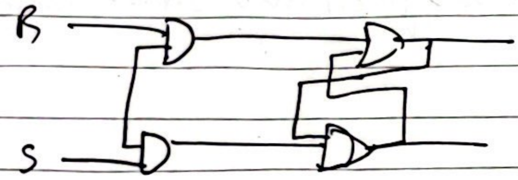
$Q(t)$ $\quad$ $Q(t+1)$
$A(t)$
$B(t)$

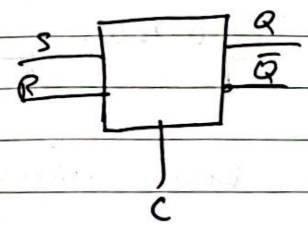
Date

set
hold/preserve
reset



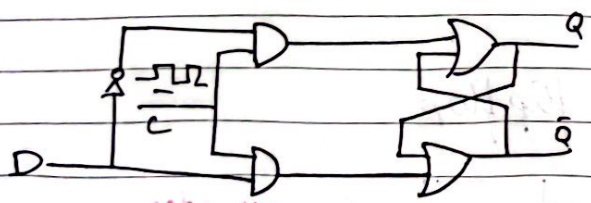
S	R	Q	$\bar{Q}$	
1	0	1	0	(set)
0	0	1	0	(hold)
0	1	0	1	(Reset)
0	0	0	1	
1	1	undefined		





$1 + X = 1$

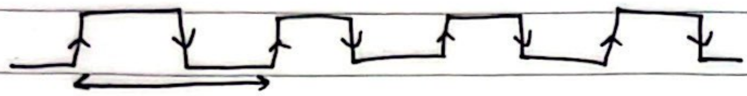
D-Latch :



Transparent

→ latches are level triggered
→ flip flops are edge triggered

Date

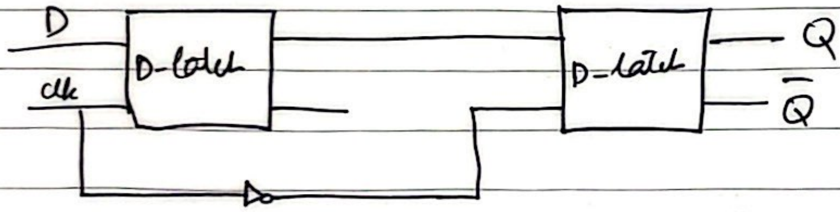


pos edge or rising
neg edge or fallaving

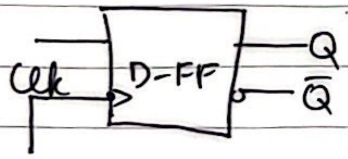
C	D	Q(t+1)
0	X	Q(t)
1	0	0
1	1	1

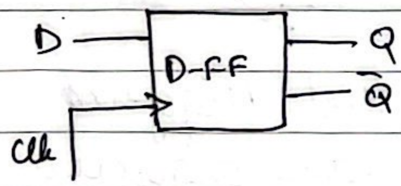
Now we will convert this level Triggered into same edge triggered circuit -

Now we are making flip flop by using latch



Negative edge triggered D-flip flop





+ve edge triggered Flip flop

Date

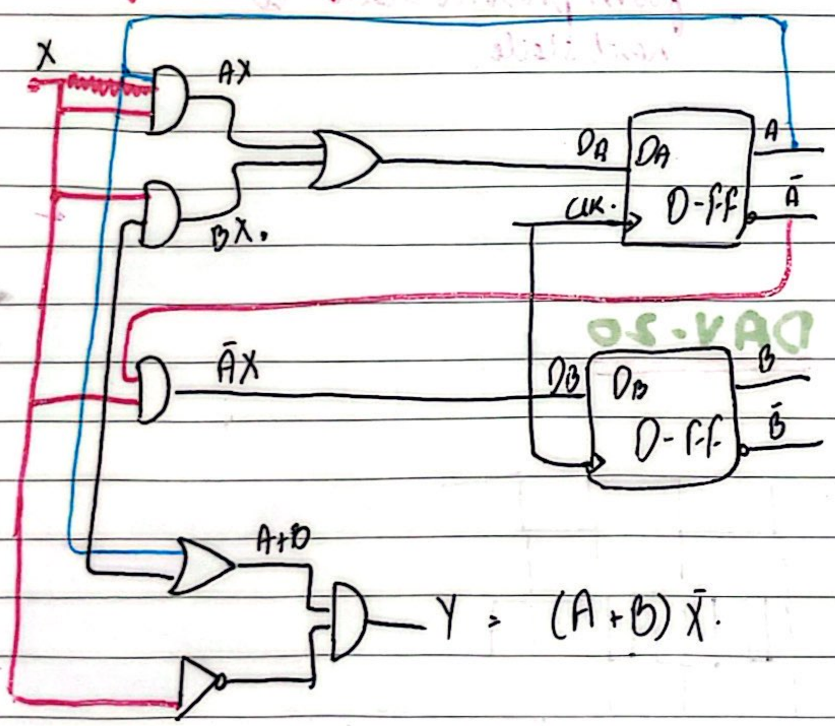
+ve edge FF

Clk	D	Q(t+1)
↑	0	0
↑	1	1
otherwise	X	Q(t)

-ve edge FF

Clk	D	Q(t+1)
↓	0	0
↓	1	1
otherwise	X	Q(t)

Analysis of Sequential Circuits:-



- → I/P
- → O/P
- → Present State
- → Next state.
- → Flip Flop Inputs.

Clk	D	Q(t+1)
↑	0	0
↑	1	1

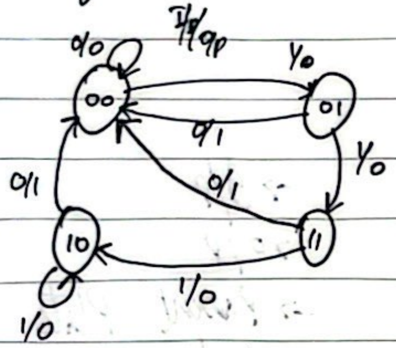
Date

Present state		Inputs	Flip Flop Inputs		Next state		Output
$A(t)$	$B(t)$	X	D_A	D_B	$A(t+1)$ D_A	$B(t+1)$ D_B	Y
0	0	0	0	0	0	0	0
0	0	1	0	1	0	1	0
0	1	0	0	0	0	0	1
0	1	1	1	1	1	1	0
1	0	0	0	0	0	0	1
1	0	1	1	0	1	0	0
1	1	0	0	0	0	0	1
1	1	1	1	0	1	0	0

State Diagram

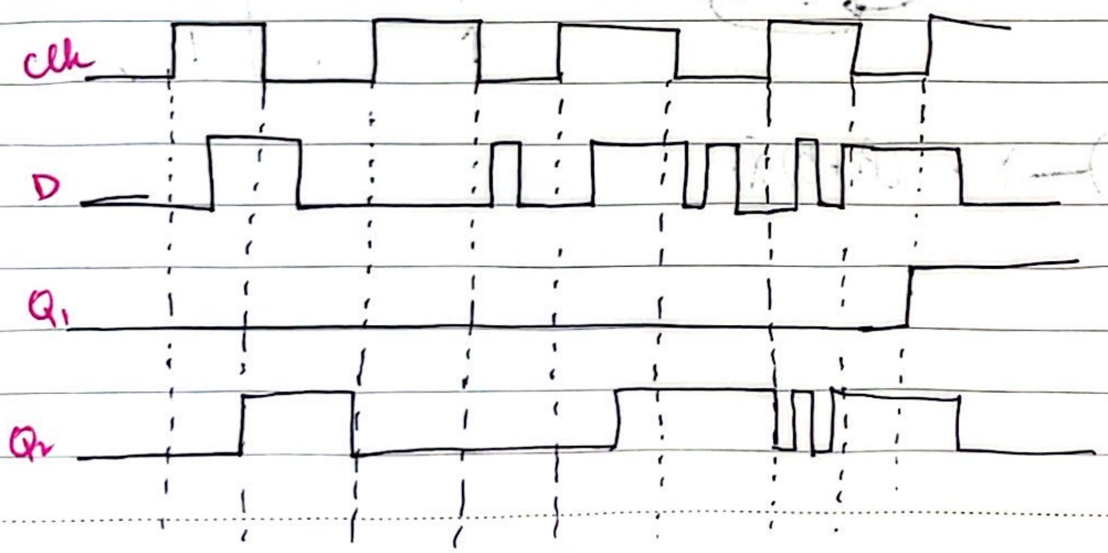
No. of circles = No. of states

No. of states = 2^n where, n = No. of FF



∴ Direction of arrow will be from present state to next state

DAY-20



Date

Mealy Model Circuits :-

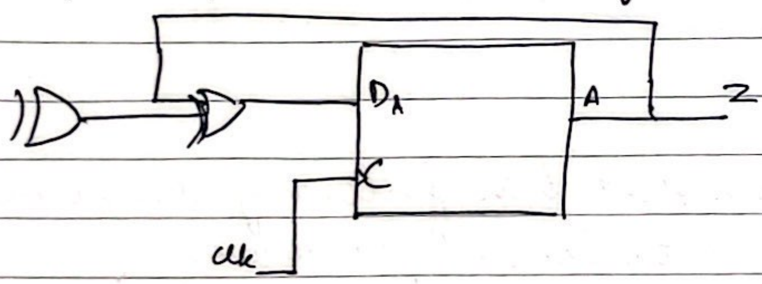
sequential circuits in which output depends on the inputs, as well as on the states -

Two dimensional state Table

Present state		Next state				Output	
		X = 0		X = 1		X = 0	X = 1
A(t)	B(t)	A(t+1)	B(t+1)	A(t+1)	B(t+1)	y	y
0	0	0	0	0	1	0	0
0	1	0	0	1	1	1	0
1	0	0	0	1	0	1	0
1	1	0	0	1	0	1	0

Moore Model Circuits :-

Output depends on states only

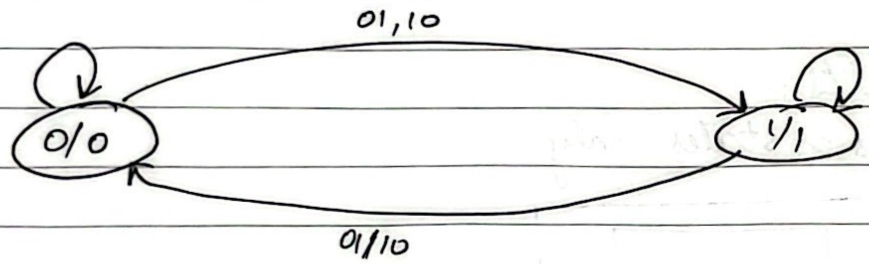


$D_A = A \oplus X \oplus Y$

Clk	D	Q(t+1)
↑	0	0
↑	1	1

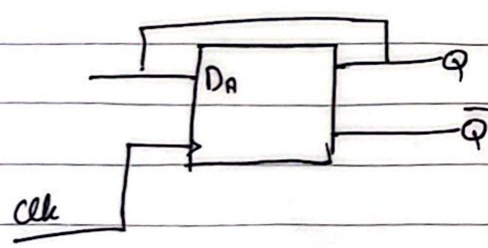
Date

Present state	Input		Next state	Output
$A(t)$	X	Y	$A(t+1)$	$Z = A$
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	0
1	0	0	1	1
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



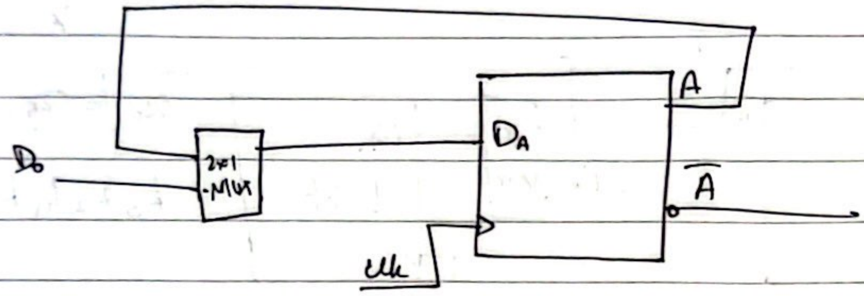
Design a digital system that has following characteristics:

1. It holds/ preserves 1 bit of data for as long as user
2. The user should have full control to set/ reset (load) the data whenever he / she wants.



Scanned with CamScanner

Date



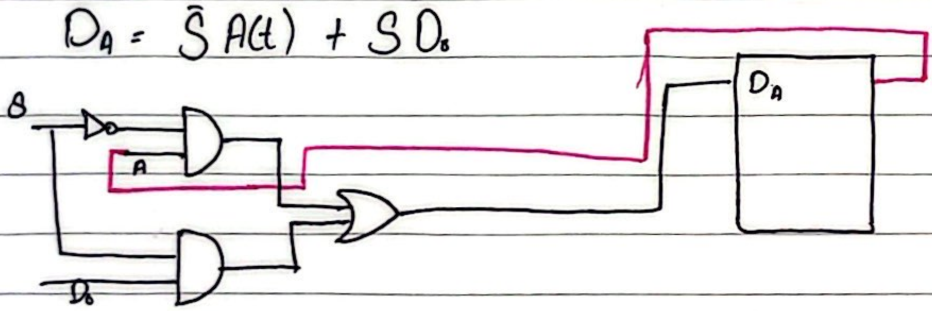
1-bit register with parallel load.

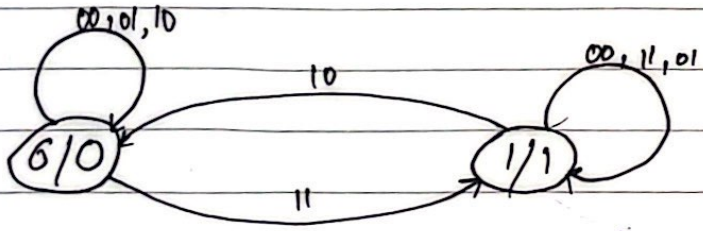
S	D_0	A(t)	A(t+1)
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

S \ D_0 A(t)	00	01	11	10
0	0	1	1	0
1	0	0	1	1

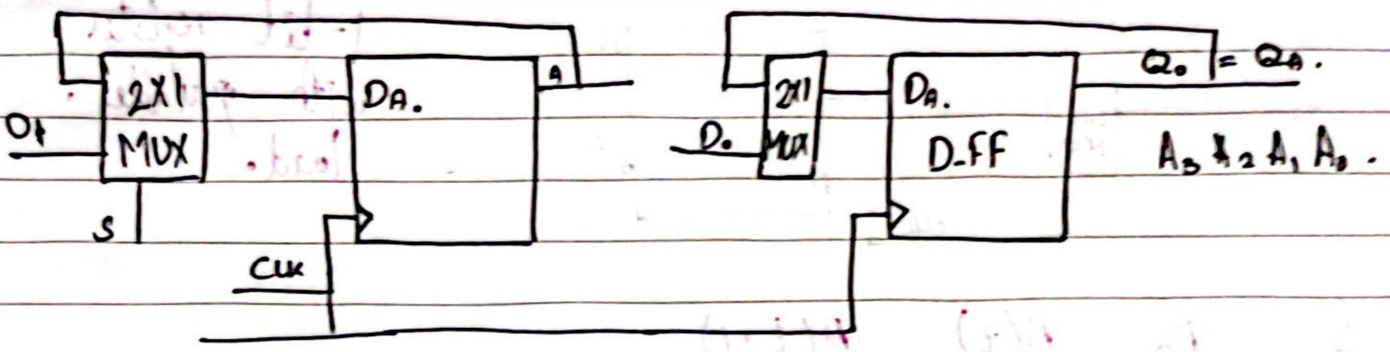
for $D_A = A(t+1)$.

$D_A = \bar{S}\,A(t) + S\,D_0$



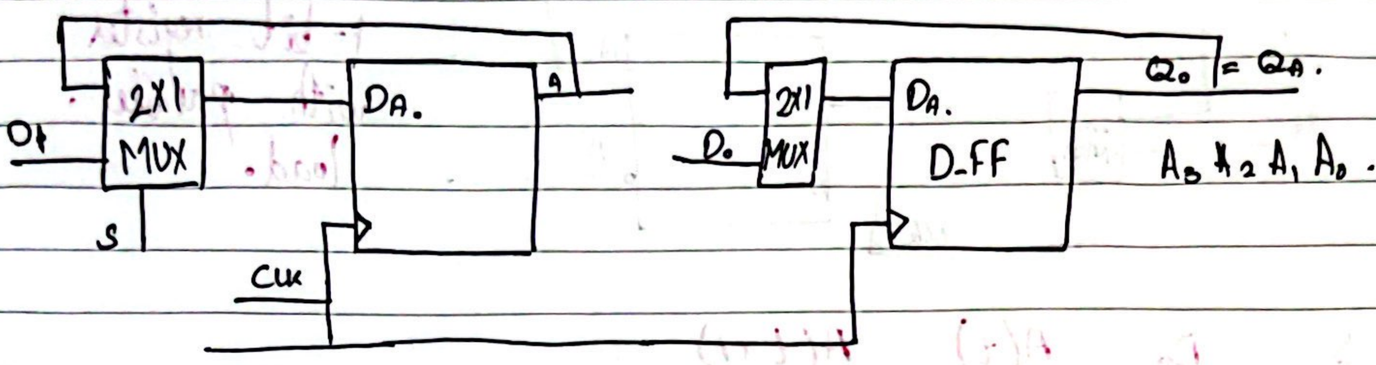


Date



S	D_1	D_0	$Q_1(t)$	$Q_2(t)$	$Q_1(t+1)$	$Q_2(t+1)$
0	X	X	0	0	0	0
0	X	X	0	1	0	1
0	X	X	1	0	1	0
0	X	X	1	1	1	1
1	0	0	X	X	0	0
1	0	1	X	X	0	1
1	1	0	X	X	1	0
1	1	1	X	X	1	1

Date



S	D1	D0	Q1(t)	Q2(t)	Q1(t+1)	Q2(t+1)
0	X	X	0	0	0	0
0	X	X	0	1	0	1
0	X	X	1	0	1	0
0	X	X	1	1	1	1
1	0	0	X	X	0	0
1	0	1	X	X	0	1
1	1	0	X	X	1	0
1	1	1	X	X	1	1

Nov 8

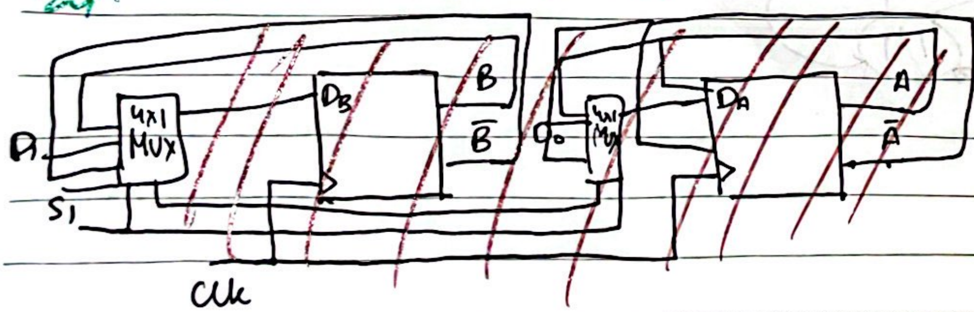
DAY-21 (online)

Question :-

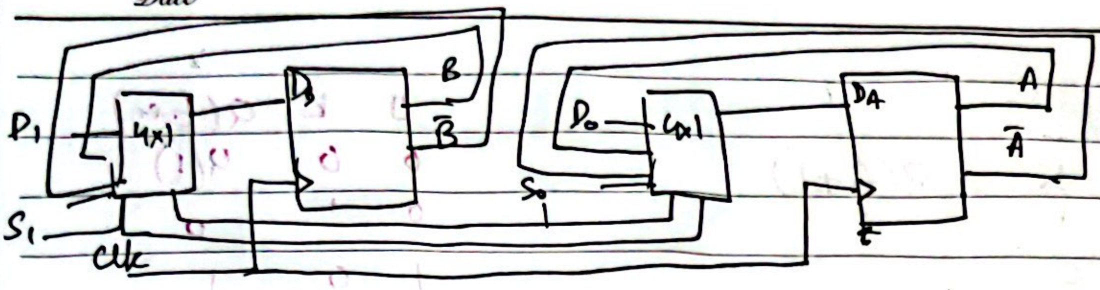
Design a digital system

1. Holds 2-bit of data
2. loads 2-bit of data
3. Toggles the data
   previous value ka complement

Sol.



Date



S_1	S_0	D_1	D_2	B(t)	A(t)	B(t+1)	A(t+1)
0	0	X	X	0	0	0	0
0	0	X	X	0	1	0	1
0	0	X	X	1	0	1	0
0	0	X	X	1	1	1	1
0	1	0	0	X	X	0	0
0	1	0	1	X	X	0	1
0	1	1	0	X	X	1	0
0	1	1	1	X	X	1	1
1	0	X	X	0	0	1	1
1	0	X	X	0	1	1	0
1	0	X	X	1	0	0	1
1	0	X	X	1	1	0	0
1	1	X	X	X	X	X	X
1	1	X	X	X	X	X	X
1	1	X	X	X	X	X	X
1	1	X	X	X	X	X	X

Characteristic table (For analysis) :-

Q(t)	D	Q(t+1)
0	0	0
0	1	1
1	0	0
1	1	1

T- flip flop

Q(t)	T	Q(t+1)
0	0	0
0	1	1
1	0	1
1	1	0

T	Q(t+1)
0	Q(t)
1	Q(t)

Date

JK Flip Flop :-

Q(t)	J	K	Q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

J	K	Q(t+1)
0	0	Q(t)
0	1	0
1	0	1
1	1	Q(w)

Excitation table :- (Design)

Q(t)	Q(t+1)	D = Q(t+1)
0	0	0
0	1	1
1	0	0
1	1	1

JK Flip flop :-

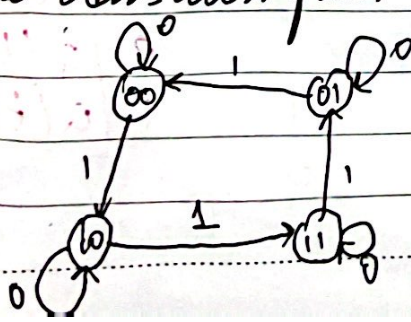
Q(t)	Q(t+1)	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

T Flip flop :-

Q(t)	Q(t+1)	T
0	0	0
0	1	1
1	0	1
1	1	0

Question :-

Design a seq. circuit using 2-D-FF and input X. X=0, the state of the system remains unchanged. X=1, circuit goes through state transition from 00 → 10 → 11 → 01 and back to 00 and repeat.



Q(t)	Q(t+1)	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Q(t)	Q(t+1)	T
0	0	0
0	1	1
1	0	1
1	1	0

Date

X	A(t)	B(t)	A(t+1)	B(t+1)	D_A	D_B	T_A	T_B	J_A	K_A	J_B	K_B
0	0	0	0	0	0	0	0	0	0	X	0	X
0	0	1	0	1	0	1	0	0	0	X	X	0
0	1	0	1	0	1	0	0	0	X	0	0	X
0	1	1	1	1	1	1	0	0	X	0	X	0
1	0	0	1	0	1	0	1	0	1	X	0	X
1	0	1	0	0	0	0	0	1	0	X	X	1
1	1	0	1	1	1	1	0	1	X	0	1	X
1	1	1	0	1	0	1	1	0	X	1	X	0

D_A

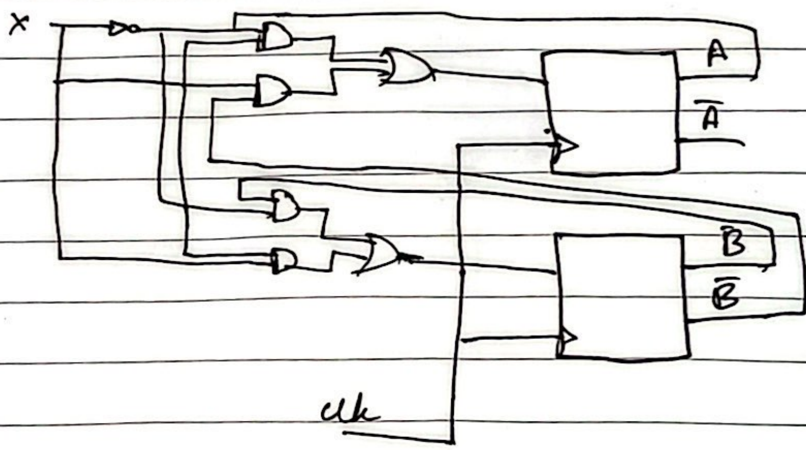
X \ A(t)B(t)	00	01	11	10
0	0	0	1	1
1	1	0	0	1

$D_A = A(t+1) = \bar{X}A(t) + X\bar{B}(t)$

D_B

X \ AB	00	01	11	10
0	0	1	1	0
1	0	0	1	1

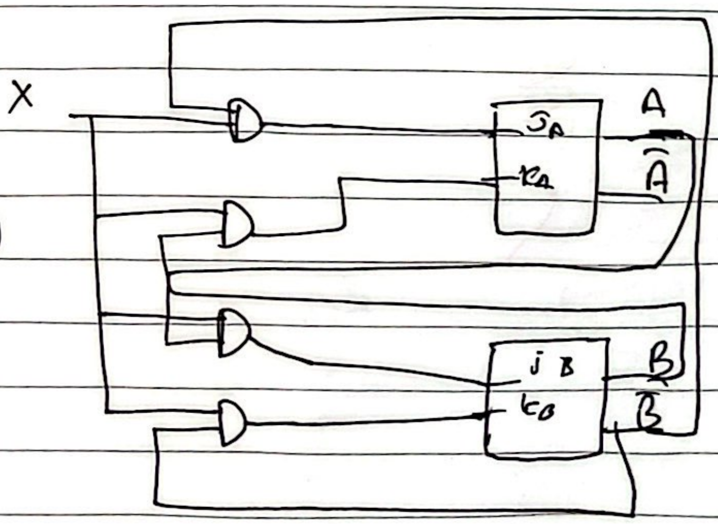
$D_B = \bar{X}B(t) + XA(t)$



X \ AB	00	01	11	10
0	0	0	X	X
1	1	0	X	X

$J_A = X\bar{B}(t)$ $\quad J_B = XA(t)$

$K_A = XB(t)$ $\quad K_B = X\bar{A}(t)$

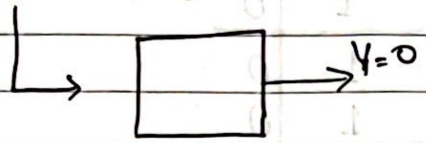


DAY-22

Design of sequence detector:

1011001110011 0001

0000 0010 000100001



3-bit sequence

$\boxed{0\ 0\ 1}$

S_0 = no bit has been recieved.

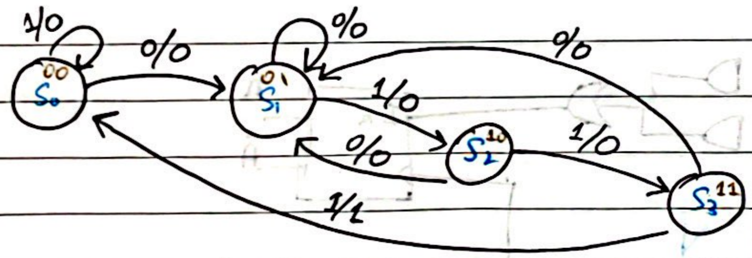
Assigning bits:-

S_0 = Initial state : No bit

S_1 = 1^{st} bit has arrived

S_2 = 2^{nd} bit has arrived

S_3 = 3^{rd} bit has arrived.



(For 0111)

Assigning codes:

S_0 = 00

S_1 = 01

S_2 = 10

S_3 = 11

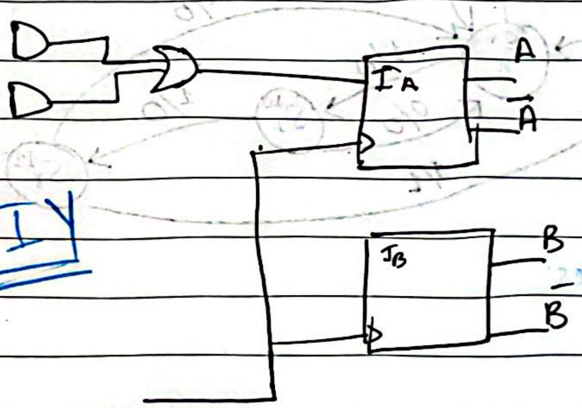
A(t)	B(t)	X	A(t+1)	B(t+1)	T_A	T_B	Y
0	0	0	0	1	0	1	0
0	0	1	0	0	0	0	0
0	1	0	0	1	0	0	0
0	1	1	1	0	1	1	0
1	0	0	0	1	1	1	0
1	0	1	1	1	0	1	0
1	1	0	0	1	1	0	0
1	1	1	0	0	1	1	1

AB \ X	0	1
00		
01		
11		
10		

A \ BX	00	01	11	10
0	0	0	1	0
1	1	0	1	1

$T_A = B(t)X + A(t)\bar{X}$

NORMAL BINARY ENCODING!!!



DIY

One Hot Encoding.

the number of flipflops are equal to the number of states.

S_0 : 1000

S_1 : 0100

Date

$S_2 = 0010$
$S_3 = 0001$

Present state:- Next state

A(t)	B(t)	C	D	X = 0	X = 1
1	0	0	0	0 1 0 0	1 0 0 0
0	1	0	0	0 1 0 0	0 0 1 0
0	0	1	0	0 1 0 0	0 0 0 1
0	0	0	1	0 1 0 0	1 0 0 0

$A(t+1) = D_A = AX + DX$
$B(t+1) = A\bar{X} + B\bar{X} + C\bar{X} + D\bar{X} = (A+B+C+D)\bar{X}$
$C(t+1) = BX$
$D(t+1) = CX$